Examiner only

5. Nuclear power in the U.K. provides around one sixth of our total electricity. It is important as it provides a constant and reliable source of electricity to help supply the base load demand for the National Grid. The incomplete nuclear equation below shows one possible fission reaction of uranium, U.

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{236}_{92}\mathrm{U} \rightarrow {}^{96}_{37}\mathrm{Rb} + {}^{\dots\dots}_{\dots\dots}\mathrm{Cs} + 3{}^{1}_{0}\mathrm{n}$$

(a) **Complete** the nuclear equation above. [2]

(b) (i) Name the particle that is absorbed by the uranium-235 nucleus. [1]

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(ii) In a nuclear reactor on average two of the three particles, $^{1}_{0}\mathrm{n}$, which are produced in one fission are absorbed by the **control rods**. Explain why this allows the nuclear reactor to operate safely. [2]

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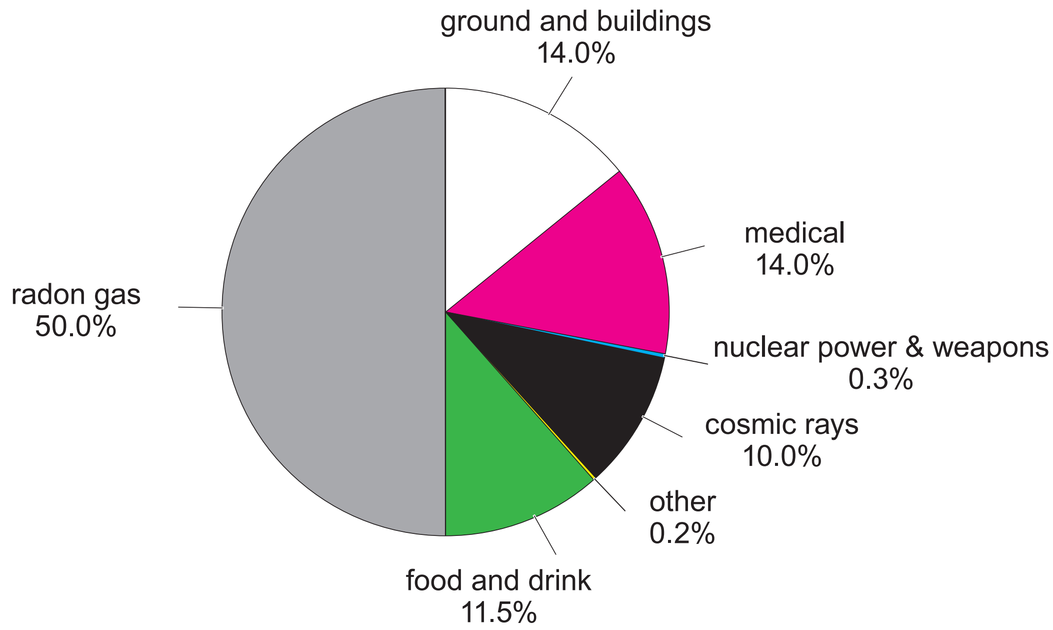
(c) The use of nuclear power is controversial and some people believe that we should not build any new nuclear power stations because of the radioactive waste that they produce. State **two** properties of nuclear waste which makes storage a problem. [2]

1. ....

2. ....

Examiner only

(d) The pie chart below shows the sources of background radiation near a nuclear power station.



(i) State how the pie chart suggests that nuclear power is a safe source of electricity. [1]

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(ii) Students measure the background radiation in counts per minute. One group takes a measurement for one minute. A second group measures the counts for 10 minutes and divide the value by 10. Explain which method is better. [2]

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